

A Minimum-Viable, Explainable Digital Twin for Dairy Cows Using Commercially Available Farm Data

Fulbright–Masaryk Postdoctoral Fellowship — Project Statement

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Proposed host: *Prof. Victor E. Cabrera, Department of Animal and Dairy Sciences, University of
Wisconsin–Madison, USA*

Proposed duration: four months

Executive Summary

A peer-reviewed survey of the 2011–2026 literature shows that an engineering-grade digital twin of an individual dairy cow does not yet exist in commercial practice. What exists are well-developed building blocks: sensors, cow-level predictive models, herd-level simulators, and decision-support workflows that have remained largely disconnected. I propose to spend four months at **University of Wisconsin–Madison** building and validating a *minimum-viable, hybrid digital twin* for dairy cows that runs on data farms already export every day — parlour / AMS logs, milk recording, management events, activity-rumination collars and weather. The twin will not attempt to replace the farmer; it will produce a daily, ranked, explainable worklist of cows to inspect, with reason codes a farmer can act on. The deliverables are a published design paper, an open prototype, a reproducible benchmark over public datasets, and a farm-deployable checklist that I will bring back to the Czech research and extension community.

Significance and Alignment with Fulbright–Masaryk Values

Dairy is among the most data-rich and most decision-poor sectors of European and U.S. agriculture. Farms generate thousands of events per day, but many small and medium farmers — on both sides of the Atlantic — still rely on memory, intuition and end-of-month spreadsheets to decide which cow to look at first. The cost is paid in unnecessary antibiotic use, premature culling, lower welfare and lost income.

Public-good framing. This project is explicitly designed so that the resulting tool can be used by family-run farms and extension veterinarians who cannot afford proprietary platforms. The open prototype, open benchmark, and farmer-facing explanation layer are the civic engagement core of the proposal — they translate state-of-the-art AI into a decision aid that respects the user’s expertise rather than overriding it.

The project advances three Fulbright–Masaryk priorities at once: (i) *scientific excellence*, by closing a documented gap between isolated alerts and value-driven decision support; (ii) *civic engagement*, by ensuring that the artefacts are open, auditable, and welfare-first; and (iii) *transatlantic transfer*, by combining U.S. leadership in herd-level decision-support tooling with Central-European strengths in operations research and mathematical modelling that I bring from ÚTIA and CULS.

State of the Art and Identified Gap

The literature is strong in monitoring and still only moderately strong in closed-loop decision-making. Sensor reviews (Rutten et al. 2013; Besler et al. 2024) and characterisation studies (Steenefeld & Hogeveen 2015) catalogue what is technically measurable and what is actually deployed. Cow-level models for mastitis, lameness, fertility, and yield perturbation (Stangaferro et al. 2016; Lemmens et al. 2023; Dervić et al. 2024; Hooker et al. 2025; Liu et al. 2025) demonstrate that activity, rumination, AMS and weather data can be combined for short-horizon risk scoring. Herd-level simulators (Calsamiglia et al. 2018; Li et al. 2023; Cabrera et al. 2026) provide what-if engines for reproduction, disease and culling. Explainability work (Mallinger et al. 2024; Guo et al. 2025; Mondini et al. 2026) has begun to translate model outputs into farmer-readable evidence.

The gap. No published system fuses these layers into one workflow that (a) runs on the data farms already export, (b) prioritises cows under realistic operator constraints, (c) explains the prioritisation in a single sentence a farmer can verify, and (d) has been validated retrospectively against alert burden, lead time, and precision-at-top- k rather than AUROC alone. Closing exactly this gap is what makes a four-month, single-host visit feasible *and* publishable.

Research Questions and Objectives

Research questions

1. What is the minimum data envelope — among milk / parlour / AMS logs, management records, activity-rumination collars and weather — that yields a useful cow-level twin?
2. Does modelling an explicit cow *state* (production state, perturbation state, fertility state, risk trajectory) outperform stand-alone one-shot predictors when the goal is prioritisation rather than raw classification?
3. How should herd-level resource constraints (operator time, veterinary slots, treatment budget) shape the cow-level workload?
4. Which explanation format (SHAP, reason codes against a cow’s own baseline, counterfactual text) produces actionable, trust-calibrated decisions for a real farmer?
5. How much of the twin’s predictive value can be demonstrated retrospectively, without a long prospective intervention trial?

Objectives

- **O1** — Define a minimum-viable digital-twin schema for a single cow and a herd.
- **O2** — Implement a hybrid prototype over retrospective farm data and public datasets.
- **O3** — Compare 2–3 baseline approaches for mastitis, lameness, fertility and yield perturbation.
- **O4** — Add an explainability layer that produces farmer-facing reason codes.
- **O5** — Add herd-level prioritisation under explicit resource constraints.
- **O6** — Deliver a public benchmark, a deployable MVP demonstrator, and a peer-reviewed design paper.

Methodology

Data sources and minimum viable stack

Three architectures are realistic for four months. Each has a clear minimum input envelope.

Architecture	Minimum inputs	Evaluation focus
Simulation-only twin	public datasets + literature parameters; no live farm API	calibration vs. literature, stability of policy, sensitivity
Hybrid twin (<i>primary</i>)	per-milking yield, parity, DIM, calvings, inseminations, treatments, activity, rumination, weather	AUROC / PR-AUC, calibration, lead time, alert burden, precision@top- <i>k</i>
Farm-deployable prototype	nightly farm export + management records + weather; collars optional	prospective utility, time saved, acceptance, false alerts

The **hybrid twin** is the primary track. Should access to a working U.S. commercial dataset through the host group materialise, the farm-deployable prototype will be a secondary deliverable; otherwise, the project falls back gracefully to open datasets from Zenodo, Agroscope, INRAE and Teagasc, all of which I have already inventoried.

Cow state representation

Each cow is represented by a latent state with four components: production state (lactation trajectory residuals), perturbation state (acute deviations in yield, conductivity, flow, rumination), fertility state (oestrus / non-oestrus / pregnant / questionable), and risk trajectory (rolling probability of mastitis / lameness / metabolic disorder). The state is updated daily; the twin emits both a point score and an uncertainty band.

Baselines and modelling stack

Task	Baseline	Stronger second step
Yield trajectory	Wood / gamma curve + mixed effects + residual anomaly score	gradient boosting over AMS + weather + collar features
Mastitis risk	penalised GLM over yield, conductivity, rumination deltas	XGBoost / LightGBM
Lameness risk	random forest over AMS / performance / mobility proxies	multimodal AMS + collar + weather
Fertility / oestrus	rules on activity spike + rumination drop + DIM filter	gradient boosting / survival model
Herd prioritisation	rule-based scoring under constraints	integer / greedy optimisation or contextual bandit
Simulation layer	Markov / state-machine + agent-based herd simulation	constrained RL once the simulator is stable

Explainability layer

- **SHAP** over tree-based models for local and global attribution.
- **Reason codes against the cow’s own baseline**, not the herd: “rumination –18% vs. 7-day mean”, “rear-quarter milk flow atypically low”, “two extra refusals”.
- **Short counterfactual sentences** of the form “if rumination were within band and milking interval shorter, this cow would drop out of today’s top 20.”
- **Uncertainty banding**: every score is rendered with a “screen now / watch / no action”

category to protect against alert fatigue.

Evaluation plan

Beyond AUROC and PR-AUC, the twin is judged on metrics that match how a farmer or veterinarian would actually use it: lead time (days before clinical diagnosis), precision@top- k (how many of the top- k flagged cows would a vet confirm?), calibration (do 0.7 alerts behave like 0.7?), and simulated workload (alerts per 100 cows per day). Where possible, evaluation is leave-one-farm-out to test transferability.

Data governance and ethics

Following the 2021 “Farmers Bill of Rights” commentary and the 2025 update on dairy data governance, every dataset will be used under an explicit licence; commercial farm data, if shared by the host group, will be anonymised at source, processed under written agreement, and never re-exported. The project will preferentially rely on open datasets (Zenodo, Agroscope, INRAE, Teagasc, data.gov.ie) so that the prototype itself can be released open source. Animal welfare is treated as the first objective, not as a constraint — the twin must never recommend an action that improves yield at the expense of welfare.

Proposed Host and Why the United States

The United States is a natural venue for this work because the U.S. dairy sector operates at the scale and degree of digital instrumentation at which the hybrid twin design becomes meaningful, and because several of the field’s most influential herd-level decision-support tools and simulators are developed at U.S. universities.

Host — Prof. Victor E. Cabrera, University of Wisconsin–Madison

Prof. Victor E. Cabrera’s group at the Department of Animal and Dairy Sciences develops web-based Markov and decision-support tools for U.S. dairy operations and has long-standing experience in dairy farm economics and herd-structure modelling. The group’s recent work (including the 2026 paper cited in this proposal) is directly adjacent to the herd-level and prioritisation layers of the proposed twin. My background in stochastic processes, Markov modelling and decision-support engineering would sit alongside that expertise as a complementary, not duplicative, contribution: I bring an open, vendor-agnostic state-estimation and explanation layer that fits on top of the herd-level tools that the Wisconsin group has already validated with U.S. producers.

Wisconsin’s Dairy Innovation Hub, its extension network, and the Department’s connections to commercial dairies in the state make UW–Madison an exceptionally well-matched environment for a four-month visit whose outputs are intended to be deployable, not only theoretical.

Why four months

Four months is a deliberately modest scope. It is long enough to deliver a working prototype, a benchmark, and a co-authored paper; short enough that the visit can fit additively into the host group’s existing roadmap rather than displace it.

Four-Month Timeline

Month	Activities	Deliverables
M1	Lock use case (mastitis / lameness / fertility triage or yield-perturbation twin); build data dictionary; implement ETL + quality-control pipeline; finalise cow-state schema.	Literature matrix, data specification, quality checklist, first twin schema.
M2	Build the yield-trajectory baseline; build 1–2 disease / risk models; validate leave-one-farm-out where data allow; produce a missingness and drift report.	Benchmark notebooks, baseline metrics, error analysis.
M3	Add the herd-level prioritisation layer under resource constraints; build SHAP / reason-code / counterfactual layer; assemble a simple dashboard or report generator; add short-horizon what-if simulations.	MVP dashboard, ranked action list, explainability layer.
M4	Compare 2–3 architectures; evaluate precision@top- k , alert burden, lead time, calibration, simulated workload; write up open limitations, governance and welfare principles; submit conference paper or preprint.	Design paper draft, open prototype repository, reproducible benchmark, farm-deployable checklist.

Deliverables and Dissemination

1. **Design paper** — *Minimum Viable Digital Twin for Dairy Cows Using Commercially Available Farm Data*, targeted at the *Journal of Dairy Science* or *Computers and Electronics in Agriculture*, with co-authorship for members of the host group secured from the outset.
2. **Open prototype** — ETL + state estimator + risk models + prioritiser + XAI layer, MIT-licensed, with reproducible notebooks.
3. **Reproducible public benchmark** — a held-out evaluation harness over Zenodo / Agroscope / INRAE / Teagasc datasets.
4. **Farm-deployable checklist** — a short document that translates the design into a step-by-step adoption guide for an extension agent or veterinarian.
5. **Outreach** — one open seminar at the host institution, one return seminar at CULS / ÚTIA, and a non-technical write-up for Czech farmer-facing media.

Risks and Mitigation

Risk	Mitigation
Commercial farm dataset unavailable in time	Project is designed to fall back to open datasets (Zenodo, Agroscope, INRAE, Teagasc, data.gov.ie) without losing deliverables.
Vendor-aggregated collar data hides preprocessing	Reason-code layer uses cow-own baselines, which are robust to vendor smoothing.
Domain shift between farms	Leave-one-farm-out evaluation is built into the pipeline from week one.
Alert fatigue in the prototype	Uncertainty banding and “no action” as a first-class output category.
Scope creep	Four-month plan locks one outcome module and one prioritisation task; everything else is explicitly a stretch goal.

Multiplier Effect for the Czech–U.S. Relationship

The stay is designed so that both sides have something durable to build on afterwards. The peer-reviewed publication, with co-authorship for members of the host group, anchors a long-term research relationship between UW–Madison and the Czech base at CULS / ÚTIA. The open prototype — hardened to a level at which a company or start-up in the United States, the Czech Republic, or jointly could adopt it as the basis of a deployment — anchors a long-term applied-research relationship. Together they translate a four-month U.S. stay into a multi-year contribution to Czech–U.S. scientific cooperation and to Czech agricultural digitalisation.

Referenced literature corresponds to the annotated bibliography included with the application; full citations are available on request.